

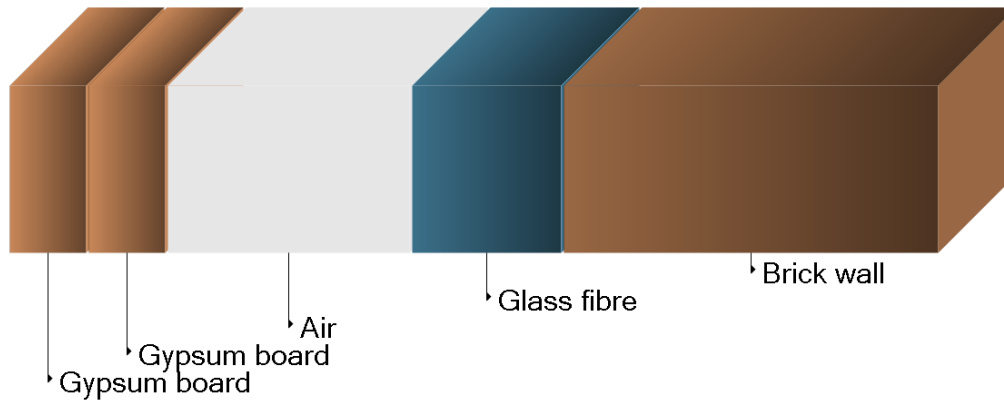
AFMG SoundFlow Report



Creation date: 16-07-2021
AFMG SoundFlow Version: 1.0.12

1. Models

1.1 Avataar - Loft - Wall 1.1- GW - Complete Assembly



Avataar - Loft - Wall 1.1- GW - Complete Assembly

Dimension	Infinite
Backing	Air
Absorber Model	Absorber
Filename	D:\PROJECTS\Avataar\Loft\Avataar - Loft - Wall 1.1- GW - Complete Assembly.sfstr

1.1.1 Layers

Overview

Layer No.	Material	Thickness [mm]	Material Type
1	Gypsum board	12.5	Plate
2	Gypsum board	12.5	Plate
3	Air	100.0	Air
4	Glass fibre	50.0	Absorber
5	Brick wall	175.0	Plate

Layer 1: Gypsum board

D.A.Bies&C.H.Hansen, Engineering Noise Control, Spon 2003, p609
Loss Factor: 0.006 - 0.03

Thickness [mm]	12.5
Material Type	Plate
Density [kg/m³]	760.0
Young's Modulus [GPa]	2.000
Poisson's Ratio	0
Bending Loss Factor	0.03

Layer 2: Gypsum board

D.A.Bies&C.H.Hansen, Engineering Noise Control, Spon 2003, p609
Loss Factor: 0.006 - 0.03

Thickness [mm]	12.5
Material Type	Plate
Density [kg/m³]	760.0
Young's Modulus [GPa]	2.000
Poisson's Ratio	0
Bending Loss Factor	0.03

Layer 3: Air

Thickness [mm]	100.0
Material Type	Air
Temperature [°C]	20.0
Relative Humidity [%]	40.0
Pressure [hPa]	1013.3

Layer 4: Glass fibre

loose

W.Fasold,E.Sonntag,H.Winkler,"Bau- und Raumakustik" Bauwesen Berlin 1987, p391

Flow Resistivity: 5 - 35 kPa*s/m²

Density: 70 - 250 kg/m³

Fibre Radius: 0.02 - 0.03 mm

Fibre Length: < 3m

Thickness [mm]	50.0
Material Type	Absorber
Flow Resistivity [kPa*s/m²]	17.0
Density [kg/m³]	140.0

Layer 5: Brick wall

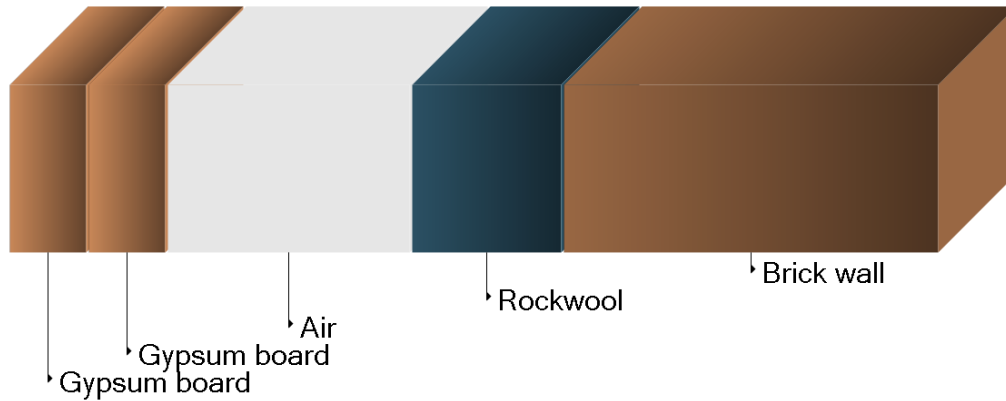
F.P.Mechel, "Formulas of Acoustics" Springer 2002, p458

bulk density: 1700 - 1800 kg/m³

E modulus: 9 - 25 GPa

Thickness [mm]	175.0
Material Type	Plate
Density [kg/m³]	1800.0
Young's Modulus [GPa]	25.000
Poisson's Ratio	0.4
Bending Loss Factor	0.04

1.2 Avataar - Loft - Wall 1.1- RW - Complete Assembly



Avataar - Loft - Wall 1.1- RW - Complete Assembly

Dimension	Infinite
Backing	Air
Absorber Model	Absorber
Filename	D:\PROJECTS\Avataar\Loft\Avataar - Loft - Wall 1.1- RW - Complete Assembly.sfstr

1.2.1 Layers

Overview

Layer No.	Material	Thickness [mm]	Material Type
1	Gypsum board	12.5	Plate
2	Gypsum board	12.5	Plate
3	Air	100.0	Air
4	Rockwool	50.0	Absorber
5	Brick wall	175.0	Plate

Layer 1: Gypsum board

D.A.Bies&C.H.Hansen, Engineering Noise Control, Spon 2003, p609
Loss Factor: 0.006 - 0.03

Thickness [mm]	12.5
Material Type	Plate
Density [kg/m³]	760.0
Young's Modulus [GPa]	2.000
Poisson's Ratio	0
Bending Loss Factor	0.03

Layer 2: Gypsum board

D.A.Bies&C.H.Hansen, Engineering Noise Control, Spon 2003, p609
Loss Factor: 0.006 - 0.03

Thickness [mm]	12.5
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Material Type	Plate
Density [kg/m³]	760.0
Young's Modulus [GPa]	2.000
Poisson's Ratio	0
Bending Loss Factor	0.03

Layer 3: Air

Thickness [mm]	100.0
Material Type	Air
Temperature [°C]	20.0
Relative Humidity [%]	40.0
Pressure [hPa]	1013.3

Layer 4: Rockwool

AFMG Example

Thickness [mm]	50.0
Material Type	Absorber
Flow Resistivity [kPa*s/m²]	45.0
Density [kg/m³]	75.0

Layer 5: Brick wall

F.P.Mechel, "Formulas of Acoustics" Springer 2002, p458

bulk density: 1700 - 1800 kg/m³

E modulus: 9 - 25 GPa

Thickness [mm]	175.0
Material Type	Plate
Density [kg/m³]	1800.0
Young's Modulus [GPa]	25.000
Poisson's Ratio	0.4
Bending Loss Factor	0.04

2. Results

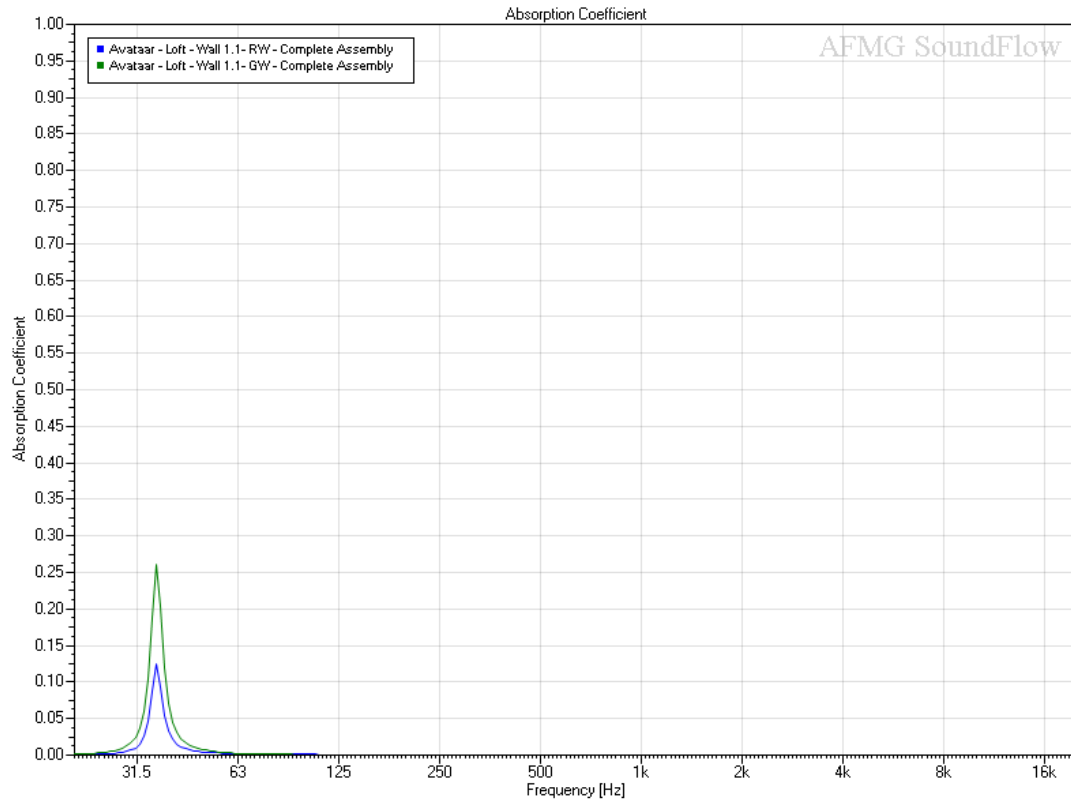
2.1 Broadband Quantities

Diffuse field incidence: 0° - 85°

Structure	ALPHA _w	NRC	R _w	C	C _{tr}	C 50-5000	C _{tr} 50-5000	STC
Avataar - Loft - Wall 1.1- GW - Complete Assembly	0.00	0.00	69	-11	-20	-13	-25	68
Avataar - Loft - Wall 1.1- RW - Complete Assembly	0.00	0.00	68	-10	-19	-12	-25	67

2.2 Absorption Coefficient

Direction of incidence	0°
Frequency resolution [Octave]	1/24
Frequency smoothing [Octave]	none



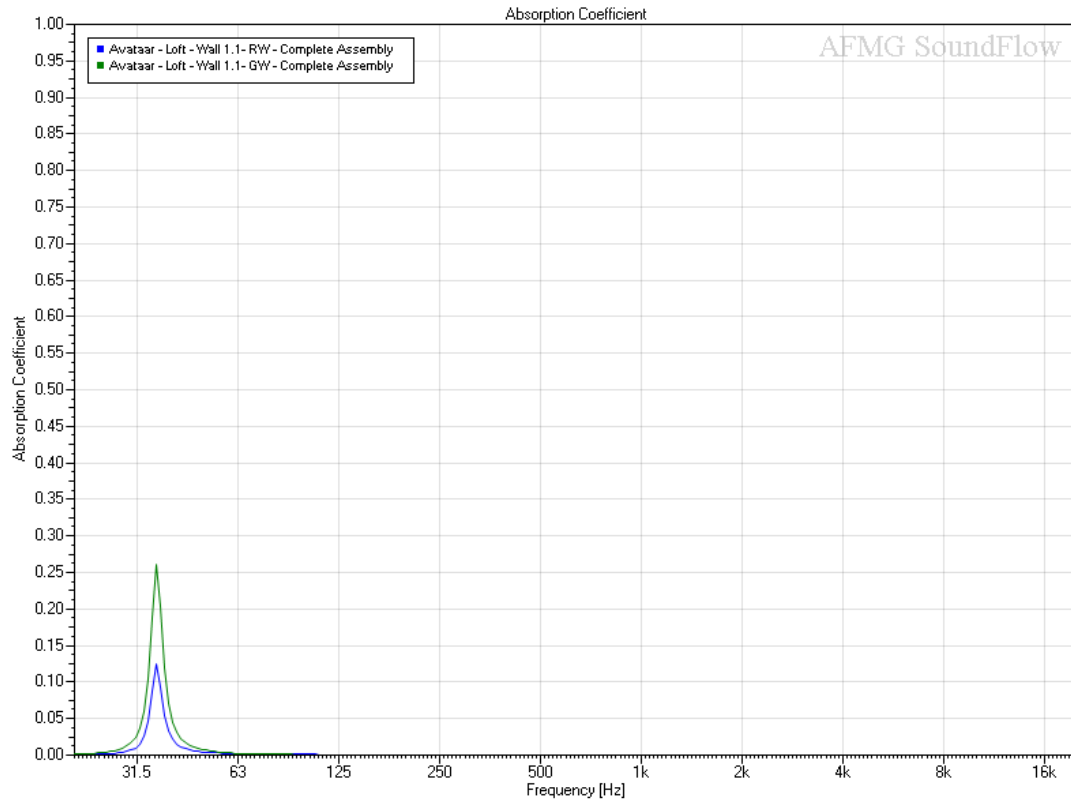
Absorption Coefficient

1/3 Octave Band Data		
Frequency [Hz]	Avataar - Loft - Wall 1.1- RW - Complete Assembly	Avataar - Loft - Wall 1.1- GW - Complete Assembly
20	0.00	0.00
25	0.00	0.00
31.5	0.03	0.07
40	0.04	0.08
50	0.00	0.01
63	0.00	0.00
80	0.00	0.00
100	0.00	0.00
125	0.00	0.00
160	0.00	0.00
200	0.00	0.00
250	0.00	0.00

315	0.00	0.00
400	0.00	0.00
500	0.00	0.00
630	0.00	0.00
800	0.00	0.00
1000	0.00	0.00
1250	0.00	0.00
1600	0.00	0.00
2000	0.00	0.00
2500	0.00	0.00
3150	0.00	0.00
4000	0.00	0.00
5000	0.00	0.00
6300	0.00	0.00
8000	0.00	0.00
10000	0.00	0.00
12500	0.00	0.00
16000	0.00	0.00
20000	0.00	0.00

2.3 Absorption Coefficient

Direction of incidence	0°
Frequency resolution [Octave]	1/24
Frequency smoothing [Octave]	none



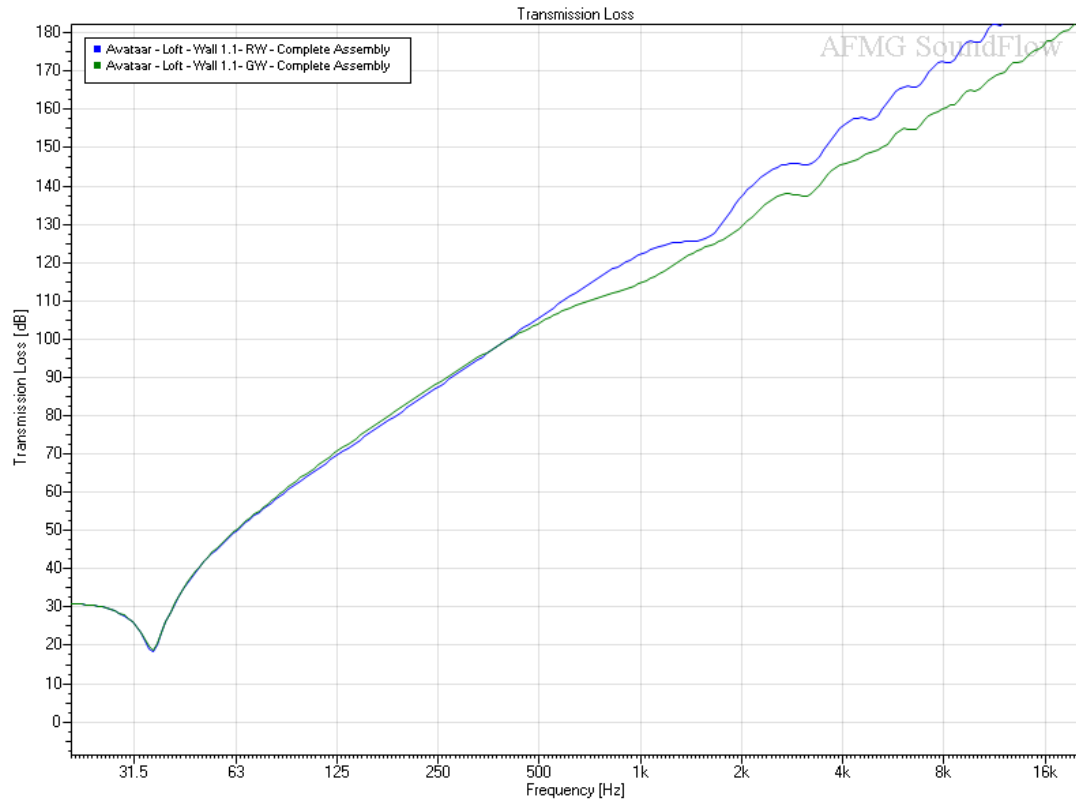
Absorption Coefficient

1/3 Octave Band Data		
Frequency [Hz]	Avataar - Loft - Wall 1.1- RW - Complete Assembly	Avataar - Loft - Wall 1.1- GW - Complete Assembly
20	0.00	0.00
25	0.00	0.00
31.5	0.03	0.07
40	0.04	0.08
50	0.00	0.01
63	0.00	0.00
80	0.00	0.00
100	0.00	0.00
125	0.00	0.00
160	0.00	0.00
200	0.00	0.00
250	0.00	0.00

315	0.00	0.00
400	0.00	0.00
500	0.00	0.00
630	0.00	0.00
800	0.00	0.00
1000	0.00	0.00
1250	0.00	0.00
1600	0.00	0.00
2000	0.00	0.00
2500	0.00	0.00
3150	0.00	0.00
4000	0.00	0.00
5000	0.00	0.00
6300	0.00	0.00
8000	0.00	0.00
10000	0.00	0.00
12500	0.00	0.00
16000	0.00	0.00
20000	0.00	0.00

2.4 Transmission Loss

Direction of incidence	0°
Frequency resolution [Octave]	1/24
Frequency smoothing [Octave]	none



Transmission Loss

1/3 Octave Band Data		
Frequency [Hz]	Avataar - Loft - Wall 1.1- RW - Complete Assembly	Avataar - Loft - Wall 1.1- GW - Complete Assembly
20	30.67	30.67
25	29.79	29.82
31.5	22.86	23.16
40	24.77	25.02
50	40.60	40.77
63	49.59	49.92
80	56.86	57.40
100	63.42	64.19
125	69.62	70.61
160	75.66	76.79
200	81.65	82.79
250	87.63	88.61

315	93.63	94.22
400	99.67	99.52
500	105.71	104.36
630	111.67	108.51
800	117.32	111.77
1000	122.18	114.83
1250	125.10	119.53
1600	127.23	124.40
2000	136.57	129.52
2500	144.50	136.72
3150	146.66	138.63
4000	154.92	145.47
5000	159.12	149.72
6300	166.18	155.25
8000	172.37	160.43
10000	178.62	165.84
12500	184.69	171.22
16000	191.67	177.07
20000	197.89	182.41